Pat Oleniuk - Software Engineer

Data | AI | Full Stack | Internal Tools | SpaceX | Netflix | CERN | Ex-People Manager

EXPERIENCE — 12 Years

Netflix — **1y, Remote, CA**
Senior Software Engineer - AI Observability — *2025*

- **Saved 3M$/year in compute and storage costs** by adding caching and reducing unnecessary data re-computation.
- Led unification of three logging libs into a one-for-all solution processing **50M events/sec, resulting in 2M$/yr savings.**
- Improved **training data quality and debuggability** by providing better **schema and lineage** and reduced obfuscation.
- Extended AI **Observability** apps to provide better **control audit of features, embeddings and datasets** (article).

SpaceX (Space flight and satellite internet provider - Starlink) — **5y, Los Angeles, CA**
Senior Software Engineer (returned to IC to build next-gen alerting) — *2025*
*Lead Software Engineer (**promoted to people manager**; led a team of 5)* — *2022*
Software Engineer II — *2021*

- Enabled **100x** customer (50k→5M+) and **10x** satellite (700→7k+) growth with only 3x more operators by building fleet automation and observability tools: aggregate/monitor from all services, alerting infra improvements, clear UIs.
- Cut **alert resolution time 5x** - led alert infrastructure rebuild involving 40 engineers, designed flexible and resilient APIs ingesting **5M events/sec**, in-stream data pipelines (Java, **Flink**, C#, **Kafka**) that stored and forwarded data into 5 systems (Postgres, Spark, Pager) providing better representation and context. The solution was later widely adopted in all programs.
- Maintain high-availability, multi-region, **real-time data products** used by 1k+ internal users **24/7/365** to operate 7k+ satellites and other fleet/network settings. Used to: update SW, control, notify on anomalies, audit and automate actions.
- Saved **$6M+/yr** by adding a new alert schema which unlocked **machine learning** engineers to train churn-reducing models.
- Drove program-wide safety and access control overhaul following a **1hr global outage** resulting in a system more robust to operator mistakes and **zero global outages** since then.
- Conducted **30+ on-call** incident investigations resulting in **hot-fixes** preventing outages and **catastrophic space collisions**.
- Saved **$500K+/yr** by co-creating a self-service change scheduling tool automating 3 full time network engineers.
- Redesigned **non-scalable** software update **tooling** to enable daily software rollouts across 5M+ devices and 20+ concurrent experiments (paper: Over-the-vacuum Update - Starlink's Approach for Upgrading Software on Thousands of Satellites).
- Boosted **productivity by 30%** by bringing 7 previously siloed teams together and reduced duplicate work.
- **Grew team from 2 to 5** multi-site direct reports, proactively coached them to achieve their goals through 1-1s, planning and standup meetings. **Promoted 4 engineers**, performance managed 1. Received excellent people manager score.
- Championed daily usage of our own apps (**'dogfooding'**) to quickly identify regressions and better understand feedback.

Virgin Hyperloop (aka Hyperloop One, mag-lev transportation startup, 300 people at-peak) — **5y, Los Angeles, CA**
Lead Software Engineer (IC tech-lead) — *2019*
Software Engineer — *2017*

- Increased simulated transportation efficiency by 70% by training & deploying **ML models** for demand prediction (Keras, Horovod, ARIMA, Prophet, Databricks, AWS) by adding event/holiday features.
- Reduced **time-to-insight 50x** by building a cloud simulation **orchestrator** and **analysis pipeline** for mass scaled simulations & grid search for generative transport designs. Tech: golang, python, postgreSQL, EC2, S3, **MLflow**, **Spark**.
- Built interactive **plots & dashboards** using plotly, DASH, bokeh, React allowing 4 other engineers deep-dive into data quicker which resulted in >20 bugfixes and AI algorithms upgrades.
- Hired and mentored 3 junior developers, presented at dev conferences: Orchestrating Mass Scaled Hyperloop Simulations.

CERN (European Org. Of Nuclear Research incl. Large Hadron Collider - LHC) — **2y, Geneva, Switzerland**
Test Software and Electronics Team Leader (IC tech-lead) — *2016*
Test Software and Electronics Engineer — *2015*

Full list of my experience on linked.in or patryk-oleniuk.github.io: Samsung, National Instruments, various ML/AI projects.

CORE SKILLS

Backend: Python, Go, C#, Java, C++, C; **APIs**: HTTP/s, protobuf, grpc; **Frontend**: HTML/CSS, Typescript, Angular, React;
Deployment: Docker, kubernetes, bazel, helm; **Observability & Reliability**: Grafana, Metabase, Prometheus, Influx;
Data Engineering: SQL, postgres, Kafka, RabbitMQ, Flink, Spark, Airflow;
Data Science: pandas, numpy, time series forecasting, signal processing, image processing, convolutional neural networks;
Leadership: hiring, performance tracking, radical candor, empathy, executive summary, vision, communication up/down.

EDUCATION

EPFL - École Polytechnique Fédérale de Lausanne — **2y, Lausanne, Switzerland**
*Masters: Electrical Engineering and Computer Science, Focus: **Data Science, Machine Learning*** — *2016-2018*

Warsaw University of Technology — **3y, Warsaw, Poland**
Bachelors: Electrical and Computer Engineering, Focus: Telecommunication and Multimedia — *2011-2015*